Lie Groups

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Abstract

Potentially good books: Humphreys, Erdmann and Wildson

Contents

1 Introduction to Lie Groups

1.1 Lie Groups

Definition 1.1.1 (Lie Groups)
A Lie group G is a group G that is also a smooth manifold such that the following are true.

- The multiplication map $\cdot : G \times G \to G$ defined by
$$(g, h) \mapsto gh$$
is a smooth map of manifolds.
- The inverse map $(-)^{-1} : G \times G$ defined by
$$g \mapsto g^{-1}$$
is a smooth map of manifolds.

Some immediate examples of Lie groups include the following:

- $(\mathbb{R}^n, +)$ and $(\mathbb{R}^n \setminus \{0\}, \times)$.
- $U(n), SU(n), O(n), SO(n), SL(n), PSL(n)$ for any $n \in \mathbb{N}$.

Example 1.1.2
Let G be a finite group equipped with the discrete topology. Then G is a Lie group.

Example 1.1.3
Let $n \in \mathbb{N}$. Then $\text{GL}(n, \mathbb{R})$ is a Lie group.

Proof
We know that $\text{GL}(n, \mathbb{R})$ is an open submanifold of $M_n(\mathbb{R}) \cong \mathbb{R}^2$. Multiplication is smooth because multiplication is given by polynomials in each entry, which are smooth. The inverse is smooth since the formula for the inverse consists of determinants of sub matrices, and they are a finite sum and product of terms of the matrices. ■

Example 1.1.4
The following are Lie groups.

- S^1.
- T^n for $n \in \mathbb{N}^+$.

Proposition 1.1.5
Let G be a Lie group. Then G is parallelizable.

Definition 1.1.6 (Lie Group Homomorphism)
Let G, H be Lie groups. Let $\phi : G \to H$ be a function. We say that f is a Lie group homomorphism if the following are true.

- f is a group homomorphism.
- f is a smooth map.

Definition 1.1.7 (Lie Group Isomorphisms)
Let G, H be Lie groups. Let $\phi : G \to H$ be a Lie group homomorphism. We say that ϕ is a Lie group isomorphism if there exists a Lie group homomorphism $\psi : H \to G$ such that
$$\psi \circ \phi = \text{id}_G \quad \text{and} \quad \phi \circ \psi = \text{id}_H$$

Lemma 1.1.8
Let G, H be Lie groups. Let $\phi : G \to H$ be a Lie group homomorphism. Then ϕ is an isomorphism if and only if ϕ is bijective.

1.2 Lie Subgroups and Quotients

Definition 1.2.1 (Lie Subgroups)
Let G be a Lie group. A Lie subgroup of G is a subset $H \subseteq G$ such that the following are true.
- H is a Lie group.
- The inclusion map $H \hookrightarrow G$ is a Lie group homomorphism.

Proposition 1.2.2
Let G be a Lie group. Let $H \leq G$ be a subgroup. If H is an embedded submanifold of G, then H is a Lie subgroup of G.

Theorem 1.2.3 (Closed Subgroup Theorem)
Let G be a Lie group. Let $H \leq G$ be a closed subgroup. Then H is a Lie subgroup of G.

Recall that any open subset of a topological group is closed. Then we have the following.

Corollary 1.2.4
Let G be a Lie group. Let $H \leq G$ be an open subgroup of G. Then H is a Lie subgroup of G.

Example 1.2.5
Let $n \in \mathbb{N}^+$. Then $\mathrm{O}(n, \mathbb{R}), \mathrm{SO}(n, \mathbb{R}), \mathrm{SL}(n, \mathbb{R})$ are Lie subgroups of $\mathrm{GL}(n, \mathbb{R})$.

Proposition 1.2.6
Let G be a Lie group. Let $H \leq G$ be a Lie subgroup. Then the following are true.
- G/H admits the structure of a smooth manifold.
- There is a vector space isomorphism
$$T_1(G/H) \cong \frac{T_1(G)}{T_1(H)}$$
- If H is normal, then G/H is a Lie group.

Theorem 1.2.7 (The First Isomorphism Theorem for Lie Groups)
Let G, H be Lie groups. Let $f : G \to H$ be a Lie group homomorphism. If $\mathrm{im}(G)$ is a closed subgroup of H, then f induces an isomorphism of Lie groups
$$\frac{G}{\ker(f)} \cong \mathrm{im}(G)$$

Let G be a topological group. Recall that G^0 is the connected component of G containing the identity.

Lemma 1.2.8
Let G be a Lie group. Then the following are true.
- $G^0 \trianglelefteq G$ is a normal subgroup.
- G^0 is a Lie group.
- G/G^0 is discrete.

2 The Lie Algebra of a Lie Group

2.1 Basic Constructions

For a group G, denote the left multiplication map of $h \in G$ by l_h. If G is a Lie group, we have seen that l_h is a smooth map, and so it induces a differential $(l_h)_*$.

Definition 2.1.1 (Left Invariant Vector Field)
Let G be a Lie group and X a vector field on G. We say that X is left invariant if

$$(l_h)_*(X_g) = X_{hg}$$

for all $X_g \in T_g(G)$.

Proposition 2.1.2
Let G be a Lie group. The vector space of left invariant vector fields of G is a Lie algebra of dimension $\dim(G)$.

Definition 2.1.3 (Lie Algebra of a Lie Group)
Let G be a Lie group. Define the Lie algebra of G to be $\mathbb{R}$-vector space

$$\text{Lie}(G) = \{X : G \to TG \mid X \text{ is a left invariant vector field } \}$$

together with the Lie bracket on vector fields.

Proposition 2.1.4
Let G be a Lie group. Then there is an $\mathbb{R}$-vector space isomorphism

$$\text{Lie}(G) \cong T_{1_G}(G)$$

given by $X \mapsto X(1_G)$.

Recall that given a homomorphism of Lie groups $\phi : G \to H$, it induces a differential $\phi_* : T_g(G) \to T_{\phi(g)}(H)$.

Proposition 2.1.5
Let G, H be Lie groups. Let $f : G \to H$ be a Lie group homomorphism. Then the induced map

$$(f_*)_{1_G} : T_{1_G}(G) \to T_{1_H}(H)$$

is a Lie algebra homomorphism.

In other words, we constructed a functor from Lie groups to Lie algebras sending a Lie group to the tangent space at the identity.

Example 2.1.6
The following are true regarding the determinant map.
- $\det : GL_n(\mathbb{R}) \to \mathbb{R}$ is a Lie group homomorphism.
- The map $(\det_*)_I : T_I(GL_n(\mathbb{R})) \to T_1(\mathbb{R})$ is given by $(\det_*)_I(X) = \text{tr}(X)$.

Example 2.1.7
Let G be a closed subgroup of $GL_n(\mathbb{R})$. Then we have

$$\text{Lie}(G) = \{M \in M_n(\mathbb{R}) \mid e^{tX} \in G \text{ for all } t \in \mathbb{R}\}$$

Proposition 2.1.8
Let G, H be Lie groups. Let $f : G \to H$ be a Lie group homomorphism. Then the following are true.

- $\text{Lie}(\ker(f)) = \ker(df)$.
- If $f(G)$ is a closed subgroup, then $\text{Lie}(\text{im}(f)) = \text{im}(df)$.

2.2 The Adjoint Map

Definition 2.2.1 (The Adjoint Map)
Let G be a Lie group. Let $\Psi_g : G \to G$ be the map $h \mapsto ghg^{-1}$. Define the adjoint map at g to be the map
$$\text{Ad}_g = (d\Psi_g)_{1_G} : T_{1_G}G \to T_{1_G}(G)$$

Lemma 2.2.2
Let G be a Lie group. Let $g \in G$. Then Ad_g is a Lie algebra homomorphism.

Example 2.2.3
Let $G \leq \text{GL}(n, \mathbb{R})$ be a closed subgroup. Let $g \in G$. Then we have
$$\text{Ad}_g(X) = gXg^{-1}$$

2.3 The Exponential Map

Definition 2.3.1 (One Parameter Subgroups)
Let G be a Lie group. A one-parameter subgroup of G is a Lie group homomorphism $\gamma : \mathbb{R} \to G$.

Lemma 2.3.2
Let G be a Lie group. Let $X \in T_{1_G}(G)$. Then there exists a unique one-parameter subgroup $\gamma : \mathbb{R} \to G$ of G such that
$$X_\gamma = X$$

Definition 2.3.3 (The Exponential Map)
Let G be a Lie group. Define the exponential map $\exp : \text{Lie}(G) \to G$ by
$$\exp(X) = \gamma(1)$$
where $\gamma : \mathbb{R} \to G$ is the unique one-parameter subgroup of G whose tangent vector at 1_G is X.

Proposition 2.3.4
Let G, H be Lie groups. Let $f : G \to H$ be a Lie group homomorphism. Let $X \in T_{1_G}(G)$. Then we have
$$\exp(f_*(X)) = f(\exp(X))$$

Recall that a finite dimensional vector space V over $\mathbb{R}$ has the structure of a manifold by identifying V with $\mathbb{R}^n$. Since $\mathbb{R}^n$ is a group, V is a Lie group whose associated Lie algebra is then $\mathbb{R}^n \cong V$.

Proposition 2.3.5
Let G be a Lie group. Then the following are true.

- $\exp : \text{Lie}(G) \to G$ is a smooth map.
- $(\exp_*)_0 : \text{Lie}(G) \to \text{Lie}(G)$ is the identity map.
- There exist open subsets $0 \in U \subseteq \text{Lie}(G)$ and $1_G \in V \subseteq G$ such that $\exp|_U$ is a diffeomorphism.

Lemma 2.3.6
Let G be an abelian Lie group. Then $\exp : \text{Lie}(G) \to G$ is a Lie group homomorphism.

Proposition 2.3.7
Let G be a Lie group. Suppose that one of the following is true.
- G is connected and compact.
- G is connected and nilpotent.
- $G = \text{GL}_n(\mathbb{R})$.

Then $\exp : \text{Lie}(G) \to G$ is surjective.

2.4 The Correspondence between Lie Groups and Lie Algebras

Theorem 2.4.1 (Lie's Third Theorem)
Let $\mathfrak{g}$ be a finite dimensional Lie algebra. Then there exists a simple connected Lie group G such that $\text{Lie}(G) \cong \mathfrak{g}$.

Theorem 2.4.2 (The Homomorphisms Theorem)
Let G, H be Lie groups such that G is simple connected. Let $\phi : \text{Lie}(G) \to \text{Lie}(H)$ be a Lie algebra homomrphisms. Then there exists a unique Lie group homomorphism $f : G \to H$ such that $\phi = df$.

Abstract nonsense: The functor F from the category of connected Lie groups to the category of finite dimensional Lie algebras admit a left adjoint Γ whose unit map $\mathfrak{g} \to F(\Gamma(\mathfrak{g}))$ is an isomorphism, and whose counit map $\Gamma(F(G)) \to G$ is the universal covering space for G.

Theorem 2.4.3 (The Subgroups-Subalgebras Theorem)
Let G be a Lie groups. Let $\mathfrak{h} \leq \text{Lie}(G)$ be a Lie subalgebra. Then there exists a unique connected Lie subgroup $H \leq G$ such that $\mathfrak{H} = \text{Lie}(H)$.

3 Smooth Actions of Lie Groups

3.1 Smooth Actions

Definition 3.1.1 (Smooth Actions)
Let G be a Lie group. Let M be a smooth manifold. An action of G on M is a function $\cdot : G \times M \to M$ such that the following are true.

- G acts on M as a group.
- $\cdot$ is a smooth map.

If G is acts on a manifold M as a Lie group, then in particular, it is also a topological group acting on a topological space continuously.

3.2 The Fundamental Vector Fields

Definition 3.2.1 (The Fundamental Vector Field)
Let M be a smooth manifold. Let G be a Lie group. Let $\Psi : G \times M \to M$ be a smooth action. Let $\xi \in \text{Lie}(G)$. Define the fundamental vector field of ξ to be the vector field $\xi_M \in \mathfrak{X}(M)$ given by

$$\xi_M(p) = \frac{d}{dt}\bigg|_{t=0} \Psi(\exp(t\xi), p)$$

Proposition 3.2.2
Let M be a smooth manifold. Let G be a Lie group. Let $\Psi : G \times M \to M$ be a smooth action. Then we have

$$\xi_M(p) = d(\Psi(-, p))_{1_G}(\xi)$$

3.3 Smooth Equivariant Maps

4 Maximal Tori

4.1 Connected Abelian Lie Groups

Proposition 4.1.1
Let G be a connected abelian Lie group. Then $\exp : \text{Lie}(G) \cong \mathbb{R}^{\dim(G)} \to G$ is the universal covering space of G.

Proposition 4.1.2
Let G be a connected abelian Lie group. Then there is a Lie group isomorphism

$$G \cong T^k \times \mathbb{R}^{\dim(G)-k}$$

for some $k \in \mathbb{N}$.

Corollary 4.1.3
Let G be a compact connected abelian Lie group. Then there is a Lie group isomorphism $G \cong T^{\dim(G)}$.

4.2 Properties of Tori

Definition 4.2.1 (Generator of a Torus)
Let $t \in T^k$. We say that t is a generator of T^k if the smallest closed subgroup containing t is T^k.

Proposition 4.2.2
Every torus admits a generator.

Proposition 4.2.3
Let $t = [(t_1, \dots, t_k)] \in (\mathbb{R}/\mathbb{Z})^k$. Then t is a generator if and only if $1, t_1, \dots, t_k$ are linearly independent over $\mathbb{Q}$.

4.3 Maximal Tori

Definition 4.3.1 (Maximal Torus)
Let G be a compact Lie group. A maximal torus of G is a Lie subgroup $T \leq G$ such that the following are true.

- T is a torus.
- T is not properly contained in a larger Lie subgroup of G that is a torus.

Proposition 4.3.2
Let G be a compact connected Lie group. Then G contains a torus.

Proposition 4.3.3
Let G be a compact connected Lie group. Let T be a maximal torus of G. Then the following are true.

- If $T' \leq G$ is another maximal torus of G, then T and T' are conjugates in G.
- We have
$$G = \bigcup_{T \text{ is a maximal torus of } G} T$$
- We have
$$Z(G) = \bigcap_{T \text{ is a maximal torus of } G} T$$

4.4 The Weyl Group

5 Representation Theory of Lie Groups

5.1 Basic Constructions

Definition 5.1.1 (Representations of Lie Groups)
Let G be a Lie group. A representation of G is a smooth map

$$\rho : G \to \text{GL}(V)$$

where V is a finite dimensional vector space over $\mathbb{R}$ and $\text{GL}(V)$ is a manifold by identifying $\text{GL}(V)$ with $\text{GL}(n, \mathbb{R})$ by choosing a basis.

Definition 5.1.2 (The Set of Representations)
Let G be a Lie group. Define the set of representation of G by

$$\text{Rep}(G) = \{\rho : G \to \text{GL}(V) \mid \rho \text{ is a representation } \}$$

Definition 5.1.3 (Homomorphism of Representations)
Let G be a Lie group. Let $\rho : G \to \text{GL}(V)$ and $\tau : G \to \text{GL}(W)$ be representations of G. A homomorphism from ρ to τ is a linear map $T : V \to W$ such that

$$T \circ \rho(g) = \tau(g) \circ T$$

for all $g \in G$.

Definition 5.1.4 (The Set of Homomorphisms of Representations)
Let G be a Lie group. Let $\rho : G \to \text{GL}(V)$ and $\tau : G \to \text{GL}(W)$ be representations of G. Define the set of homomorphisms from ρ to τ by

$$\text{Hom}_{\text{Rep}(G)}(V, W) = \{T : V \to W \mid T \text{ is a homomorphism of representations } \}$$

Proposition 5.1.5
Let G be a Lie group. Let $\rho : G \to \text{GL}(V)$ be a representation of G. Then $\rho_* : \text{Lie}(G) \to \mathfrak{gl}(V)$ is a representation of Lie algebras.

Proposition 5.1.6
Let G be a simply connected Lie group. Then the following are true.

- There is a natural bijection
$$\text{Rep}(G) \quad \overset{1:1}{\longleftrightarrow} \quad \text{Rep}(\mathfrak{g})$$
- There is a natural bijection
$$\text{Hom}_{\text{Rep}(G)}(V, W) \quad \overset{1:1}{\longleftrightarrow} \quad \text{Hom}_{\text{Rep}(\text{Lie}(G))}(V, W)$$

6 Homotopy Theory of Lie Groups

Proposition 6.0.1
Let G be a connected Lie group. Then the following are true.
- G admits a universal covering space $\widetilde{G}$ that is a Lie group.
- The structure map $p : \widetilde{G} \to G$ is a morphism of Lie groups.
- We have $\pi_1(G) \cong \ker(p)$.

Example 6.0.2
The universal covering space of T^n is $\mathbb{R}^n$, which is also a Lie group and whose covering map is a Lie group homomorphism.

Example 6.0.3
The map $\exp : \mathbb{R} \to \mathbb{R}^*$ given by $t \mapsto e^t$ is a universal covering map of $\mathbb{R}^*$.